Ps 04

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1 Problem 36

1.1 Task

> Professor Gallup wants to create a random sample of the set $\{1, \ldots, n\}$, that is, an m element subset S, where $1 \leq m \leq n$, such that each m subset is equally likely to be created. One way is to set $A[i] = i$, for $i = 1, 2, \ldots, n$, call RANDOMLY-PERMUTE(A), and then take just the first m array elements. Show that this indeed produces a random sample.

1.2 Solution

Fix $T \subseteq \{1, \ldots, n\}$ with $|T| = m$.

RANDOMLY-PERMUTE yields each permutation with probability $1/n!$. Number of permutations whose first m positions contain exactly T:

- arrange T in positions $1, \ldots, m$: $m!$ ways,
- arrange $\{1, \ldots, n\} \setminus T$ in positions $m+1, \ldots, n$: $(n-m)!$ ways.

$$\Pr[S = T] = \frac{m!(n-m)!}{n!} = \frac{1}{\binom{n}{m}}.$$

Independent of T, so all $\binom{n}{m}$ subsets are equiprobable. ■

1.2.1 Numerical check

For $n = 10$, $m = 3$: $\Pr[S = T] = \frac{3!\,7!}{10!} = \frac{1}{\binom{10}{3}} = \frac{1}{120} \approx 0.00833$.

```
using Random
let
    n, m = 10, 3
    target = Set(1:m)               # fix an arbitrary 3-subset
    epochs = 1_000_000
    hits = 0
    for _ in 1:epochs
        sample = Set(shuffle(1:n)[1:m])
        hits += (sample == target)
    end
    hits / epochs                   # should be  1/120  0.00833
end
```

2 Problem 37

2.1 Task

The method from Exercise 36 makes n calls to the RANDOM procedure. In Professor Gallup's application, n is much larger than m, and so the professor wants to create a random sample with fewer calls to RANDOM.

1: procedure RANDOM SAMPLE(m, n) 2: $S = \emptyset$ 3: for $k = n - m + 1$ to n do 4: $i = \text{RANDOM}(1, k)$ 5: if $i \in S$ then 6: $S = S \cup \{k\}$ 7: else 8: $S = S \cup \{i\}$ 9: end if 10: end for 11: return S 12: end procedure

Show that the procedure RANDOM SAMPLE returns a random m subset S of $\{1, \ldots, n\}$.

2.2 Solution

Don't really want to do this one...

3 Problem 38

3.1 Task

Professor Knievel suggests the procedure PERMUTE-BY-CYCLE to generate a uniform random permutation. Show that each element $A[i]$ has a $1/n$ probability of winding up in any particular position in B. Then show that Professor Knievel is mistaken by showing that the resulting permutation is not uniformly random.

1: procedure PERMUTE-BY-CYCLE(A, n) 2: let $B[1 : n]$ be a new array 3: offset $=$ RANDOM$(1, n)$ 4: for $i = 1$ to n do 5: dest $= i +$ offset 6: if dest $> n$ then 7: dest $=$ dest $- n$ 8: end if 9: $B[\text{dest}] = A[i]$ 10: end for 11: return B 12: end procedure

3.2 Solution

3.2.1 (a)

For a fixed i and target position j, $A[i]$ lands at j iff $i + \text{offset} \equiv j \mod n$. Exactly one offset in $\{1, \ldots, n\}$ satisfies this. Since the offset is uniform on $\{1, \ldots, n\}$, probability is $\frac{1}{n}$.

3.2.2 (b)

The algorithm is fully determined by the choice of offset $\in \{1, \ldots, n\}$, so it produces at most n distinct permutations. But $\{1, \ldots, n\}$ admits $n!$ permutations, and $n < n!$ for $n \geq 3$. Hence at least $n! - n$ permutations occur with probability 0, so the distribution is not uniform.

4 Problem 39

4.1 Task

Professor Kelp decides to write a procedure that produces at random any permutation except the identity permutation, in which every element ends up where it started. He proposes the procedure PERMUTE-WITHOUT-IDENTITY. Does this procedure do what Professor Kelp intends?

1: procedure PERMUTE-WITHOUT-IDENTITY(A, n) 2: for $i = 1$ to $n-1$ do 3: swap $A[i]$ with $A[\text{RANDOM}(i+1, n)]$ 4: end for 5: end procedure

4.2 Solution

4.2.1 (a)

Yes. At iteration $i = 1$, the algorithm swaps $A[1]$ with $A[\text{RANDOM}(2, n)]$, so $A[1]$ is displaced. Hence position 1 cannot retain its original element, and the identity permutation is never produced.

4.2.2 (b)

No — the distribution is not uniform over the $n! - 1$ non-identity permutations.

The algorithm performs $n - 1$ iterations; iteration i draws from a range of size $n - i$. Total number of outcome sequences:

$$\prod_{i=1}^{n-1}(n-i) = (n-1)!.$$

For $n \geq 3$, $(n-1)! < n! - 1$, so at least $n! - 1 - (n-1)!$ non-identity permutations are unreachable and occur with probability 0. ■

5 Problem 40

5.1 Task

Consider the PERMUTE-WITH-ALL, which instead of swapping element $A[i]$ with a random element from the subarray $A[i : n]$, swaps it with a random element from anywhere in the array. Does PERMUTE-WITH-ALL produce a uniform random permutation? Why or why not?

1: procedure PERMUTE-WITH-ALL(A, n) 2: for $i = 1$ to n do 3: swap $A[i]$ with $A[\text{RANDOM}(1, n)]$ 4: end for 5: end procedure

5.2 Solution

No. The algorithm runs n iterations, each independently choosing an index from $\{1, \ldots, n\}$, giving n^n equally likely execution paths, each with probability $1/n^n$.

If the resulting distribution over the $n!$ permutations were uniform, each permutation would be produced by exactly $n^n/n!$ paths. But the number of paths producing a given permutation is a non-negative integer, so uniformity requires $n! \mid n^n$.

For $n = 3$: $n^n/n! = 27/6 = 9/2 \notin \mathbb{Z}$, so $3! \nmid 3^3$. Hence the distribution is not uniform. ■

6 Problem 41

6.1 Task

What are the minimum and maximum numbers of elements in a heap of height h? Show that an n element heap has height $\lfloor \lg n \rfloor$.

6.2 Solution

6.2.1 (a)

Maximum number - $2^{h+1} - 1$, minimum - 2^h.

6.2.2 (b)

A heap of height h with n elements satisfies, by (a),

$$\begin{aligned} 2^h &\leq n \leq 2^{h+1} - 1 \\ h &\leq \lg n \leq \lg(2^{h+1} - 1) < h + 1 \\ h &\leq \lg n < h + 1 \end{aligned}$$

Hence $h = \lfloor \lg n \rfloor$. ■

7 Problem 42

7.1 Task

Consider a max-heap, in which all elements are distinct. Where can the smallest element reside? At which levels in a max-heap might the k th largest element reside, for $2 \leq k \leq \left\lfloor \frac{n}{2} \right\rfloor$?

7.2 Solution

7.2.1 (a)

Leaves. In a max-heap any internal node exceeds its children, so the minimum cannot have any. Array indices $\lfloor n/2 \rfloor + 1, \ldots, n$.

7.2.2 (b)

$$1 \leq \ell \leq \min(k-1, \lfloor \lg n \rfloor).$$

Lower: $k \geq 2$ excludes the root.

Upper: the ℓ strict ancestors of the k th largest are each greater than it, so $\ell \leq k-1$; combined with the heap-height bound $\ell \leq \lfloor \lg n \rfloor$. ■

8 Problem 43

8.1 Task

Is an array that is in sorted order a min-heap? Is the array with values

$$[33, 19, 20, 15, 13, 10, 2, 13, 16, 12]$$

a max-heap?

8.2 Solution

8.2.1 (a)

Yes. For sorted ascending A and any valid i: $i < 2i$ and $i < 2i+1$, so $A[i] \leq A[2i]$ and $A[i] \leq A[2i+1]$. Min-heap property holds. Converse fails (a min-heap need not be sorted).

8.2.2 (b)

No. $A[4] = 15$ but $A[9] = 16$, violating $A[i] \geq A[2i+1]$.

```
let
    arr = [33, 19, 20, 15, 13, 10, 2, 13, 16, 12]
    n = length(arr)
    ok = true
    for i in 1:div(n, 2)
        if arr[i] < arr[2i] || (2i+1  n && arr[i] < arr[2i+1])
            ok = false
            break
        end
    end
    println(ok ? "max-heap" : "not max-heap")
end
```